

A pre-existing CloudTrail trail is stopped and deleted, or its log-delivery bucket is set to expire objects within an abnormally short retention window

Incident 6bc323a620a6443f · Generated Aug 21, 2026, 11:47:40 AM UTC

HIGH SEVERITY	35% MATCH CONFIDENCE	ESCALATED FOR INVESTIGATION	VERDICT: SUPPORTED (80%)
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Overview

An alert detected a CloudTrail logging stop event initiated by the `northwindsso-engineering/dana.whitfield` role, escalating due to its corroboration with preceding cloud account discovery activity. The Tier 3 investigation concluded with high confidence that this activity is indicative of an attacker attempting to cover their tracks by disabling logging after performing enumeration within the environment. The hypothesis of legitimate administrative or maintenance activity was rejected due to the absence of corresponding re-enablement or trail creation events and the lack of a covering change ticket. With an overall confidence of 0.8, the evidence strongly points towards a malicious defense evasion attempt.

Timeline

Aug 15, 2026, 11:22:40 AM UTC Detection rule R-T1685.002-NET-001 matched (confidence 0.35)

Platform Analysis

s1	event_type_in: ['StopLogging', 'DeleteTrail']
q1	Not confirmed: Was a replacement trail created to cover the same account/region before or immediately after this one was removed?
q2	Not confirmed: Does this account have a documented log-retention policy that requires this trail's audit history to be preserved?

MITRE ATT&CK Mapping

Technique	T1685.002
Tactic	defense-evasion
Description	A pre-existing CloudTrail trail is stopped and deleted, or its log-delivery bucket is set to expire objects within an abnormally short retention window

Correlation Analysis

R-T1685.002-NET-001 (root detection)

Preceding: discovery	Within 30m · iam_principal: arn:aws:sts::210987654321:assumed-role/northwindsso-engineering/dana.whitfield
Preceding: privilege-escalation	Within 2h · iam_principal: arn:aws:sts::210987654321:assumed-role/northwindsso-engineering/dana.whitfield
Following: collection	Within 2h · iam_principal: arn:aws:sts::210987654321:assumed-role/northwindsso-engineering/dana.whitfield

Following: exfiltration

Within 4h · iam_principal:
arn:aws:sts::210987654321:assumed-role/northwindsso-engineering/dana.whitfield

R-T1087.004-NET-002 (corroborating detection)

No related activity was found.

Disposition

This alert was corroborated by 1 related event(s) elsewhere in this environment -- a connected chain of activity, not an isolated alert.

Tier 3 Investigation

Trigger	This alert was corroborated by 1 related event(s) elsewhere in this environment -- a connected chain of activity, not an isolated alert.
Finding	The investigation reveals a strong likelihood of malicious activity. An AWS IAM principal, `dana.whitfield`, operating under the `northwindsso-engineering` assumed role, stopped CloudTrail logging. This action was immediately preceded by `Cloud Account Discovery` activity from the same principal. There is no evidence of a replacement CloudTrail being created, logging being re-enabled, or any documented change management window covering this activity. This pattern strongly suggests an attacker attempting to obscure their actions after performing reconnaissance within the AWS environment. The `source_ip` (198.51.100.77) associated with the `StopLogging` event should be further investigated.
Verdict	Hypothesis H1 was supported (80% confidence)

Hypotheses Considered (2)

SUPPORTED	H1 · confidence 80%
The `StopLogging` event is indicative of an attacker attempting to cover their tracks by disabling CloudTrail logging to evade detection. The correlated `R-T1087.004-NET-002` (Account Discovery: Cloud Account) further supports this by suggesting the attacker performed enumeration activities before or during the logging disablement. Supporting evidence: ANCHOR EVIDENCE: The `StopLogging` event directly aligns with the tactic of defense evasion (T1685.002).; RECURSIVE FOLLOW-UP INVESTIGATION: `R-T1087.004-NET-002` fired, indicating Cloud Account discovery activity by the same principal (`arn:aws:sts::210087654321:assumed-role/northwindsso-engineering/dana.whitfield`) around the same time.; get_identity_verdict_history: The same IAM principal `dana.whitfield` has a verdict for `R-T1087.004-NET-002` (Account Discovery: Cloud Account) with an `observed_start` time (`2026-08-15T11:20:03+00:00`) that precedes the `StopLogging` event (`2026-08-15T11:22:40+00:00`), suggesting an enumeration phase before attempting to hide activity.	
REJECTED	H2 · confidence 20%
The `StopLogging` event is part of a legitimate administrative or maintenance activity. The subsequent or preceding discovery activity (R-T1087.004-NET-002) could also be part of routine checks or automation within the 'northwindsso-engineering' role, and the lack of a replacement trail or re-enabled logging could be due to a temporary suspension or a planned change not captured by available telemetry. Disconfirming evidence: search_alert_pool (for CreateTrail/StartLogging): No evidence of a replacement CloudTrail being created or logging being re-enabled around the anchor time, which would typically follow a legitimate suspension.; get_cloud_resource_activity: No CloudTrail-related API calls by the principal around the anchor time other than the 'StopLogging' event, which does not suggest legitimate re-configuration or re-enabling.; check_context_store (active_change_ticket_covering_asset): No active change ticket was found to cover the activity, reducing the likelihood of planned maintenance. Rejected because: The absence of any corresponding 'CreateTrail' or 'StartLogging' events, combined with the lack of an active change ticket covering the activity, significantly weakens the hypothesis of a legitimate administrative action. While the tools couldn't definitively prove the absence of <i>any</i> planned change, the available evidence does not support a benign explanation, especially when paired with preceding account discovery activity.	

Recommended Follow-up

- Investigate the source IP `198.51.100.77` for reputation, geolocation, and any other activity within the environment, although no automated response is configured for this detection; isolating this host would require a manual, out-of-band action.
- Inspect the `northwindsso-engineering` IAM role's permissions to understand its full scope and identify any excessive privileges.
- Review CloudTrail logs for `dana.whitfield`'s activity *prior* to the `StopLogging` event for other suspicious actions that might not have triggered specific detection rules, specifically looking for `List` or `Describe` API calls.

False Positive Analysis

Fail: Is the acting process/tool/principal one that is expected or documented for this environment (not spoofed, not anomalous for this asset/user)?	While the principal `dana.whitfield` under `northwindsso-engineering` might be an expected actor in the environment, the specific action of stopping CloudTrail logging is not expected for routine operations, especially when coupled with prior discovery activity and no corresponding change ticket. No evidence suggests this is an authorized or routine action for this identity.
Fail: Does the specific execution/access pattern observed match ordinary, benign usage, rather than a known malicious technique shape?	The combination of Cloud Account Discovery followed by stopping CloudTrail logging does not match an ordinary, benign usage pattern. This sequence is a known malicious technique for defense evasion.
Fail: Is this activity covered by a documented change ticket, maintenance window, or pentest-calendar entry?	No active change ticket covering the anchor time was found for the asset or principal. No pentest calendar entry was found either, indicating the activity is not covered by documented authorization.
Fail: Is this alert corroborated by at least one independent piece of evidence (a resolved qualifier, a real correlated hop, or additional tool-call evidence), rather than resting on a single uncorroborated detection?	While `get_corroborating_entities` did not return other corroborating verdicts, `get_identity_verdict_history` shows a related `R-T1087.004-NET-002` (Account Discovery) verdict for the same principal immediately preceding the `StopLogging` event. This internal correlation strengthens the malicious hypothesis, even though it wasn't a separate, distinct verdict in `get_corroborating_entities`.

Detection Opportunities

T1685.002	Enhance the existing detection for `StopLogging` to include a required qualifier that checks for subsequent `CreateTrail` or `StartLogging` events within a short timeframe (e.g., 5-10 minutes) by the same principal, with a higher severity if no such event is found.
T1087.004	Create a detection rule that specifically looks for `Cloud Account Discovery` (T1087.004) events followed by `StopLogging` (T1685.002) or `DeleteTrail` within a short window (e.g., 10 minutes) by the same IAM principal, as this represents a strong attack chain.
T1087.004	Implement baselining for `List/Describe` API calls made by IAM principals. Alert on significant deviations from normal patterns, especially if an assumed role, typically used by automation, begins performing broad enumeration.

Lessons Learned

- The absence of expected benign activity (e.g., re-enabling logging, change ticket) can be as strong an indicator of malicious intent as the presence of outright malicious activity.
- Understanding the sequence of events is critical: initial reconnaissance (T1087.004) followed by defense evasion (T1685.002) is a common attacker pattern that should be explicitly linked in detection logic.
- The platform's `get\_corroborating\_entities` tool is useful for broad cross-platform correlations, but internal correlations within the same platform (e.g., CDR) on the same entity (IAM principal) also provide valuable context and can be surfaced through `get\_identity\_verdict\_history`.

Evidence Gathered

10 queries executed during the investigation · 7 record(s) reviewed

`get_corroborating_entities`

`get_detection_dependencies`

search_alert_pool	sensor: cdr, same_asset_only: True, event_type: CreateTrail, within_seconds_of_anchor: 1800
search_alert_pool	event_type: CreateTrail, same_asset_only: True, sensor: cdr, within_seconds_of_anchor: 1800
search_alert_pool	event_type: StartLogging, same_asset_only: True, within_seconds_of_anchor: 1800, sensor: cdr
get_cloud_resource_activity	resource_name: arn:aws:sts::210987654321:assumed-role/northwindsso-engineering/dana.whitfield
check_context_store	check: active_change_ticket_covering_asset
check_context_store	check: pentest_calendar
get_identity_verdict_history	
get_asset_verdict_history	

Investigation duration: 73.0s · 10 question(s) asked · 7 record(s) analyzed

Recommended Actions

No actions were warranted at this confidence level.